AVASH ARYAL

San Marcos, TX | 512-300-5589 | aryalavash122@gmail.com
www.linkedin.com/in/avasharyal | avash-a.github.io/Personal_Website/

EDUCATION

Texas State University | San Marcos, TX Expected May 2027

Bachelor of Science in Electrical Engineering

Concentration: Micro and Nano Devices and Systems

Minor: Applied Mathematics

Engineering GPA: 3.97/4.00 | Honors College | Dean's List

RELEVANT COURSEWORK

- Introduction to VLSI Design (In Progress)
- Solid State Devices
- Microelectronics
- Microelectronics Manufacturing
- Circuits I & II
- Digital Logic
- Microprocessors
- Electromagnetics
- Signals and Systems
- Linear Algebra

RESEARCH EXPERIENCE

Undergraduate Semiconductor Researcher Spring 2026 - Present

Texas State University | Research with Dr. Dinusha

- Conducted semiconductor device research using Silvaco TCAD (ATLAS), investigating AlScN-based ferroelectric structures and emerging wide-bandgap semiconductor devices.
- Modeled ferroelectric polarization and electrical behavior, including hysteresis and the influence of device and material parameters.
- Investigated simulated I–V and C–V characteristics, including drain-current versus gate-voltage and drain-current versus drain-voltage relationships.
- Analyzed relationships among semiconductor device physics, ferroelectric material properties, and electrical performance.

SELECTED ENGINEERING PROJECTS

GaN Device Fabrication and Hybrid Renewable-Energy Power Conversion

Project Manager & Power Electronics Lead | Senior Design

Texas State University Fall 2026 - Spring 2027 | Ongoing

- Lead a team developing a hybrid solar–wind energy harvesting and battery-charging system; coordinate technical responsibilities, milestones, and communication with the project sponsor.
- Collaborate with team members on an ongoing GaN semiconductor device fabrication effort for intended use in power-conversion hardware.
- Design, simulate, prototype, and evaluate buck and boost DC–DC converters for renewable-energy power conditioning and LiFePO4 battery charging.
- Develop switching circuits, gate-driver interfaces, and PCB designs using LTspice and Altium Designer, with laboratory instrumentation supporting hardware validation.

SELECTED ENGINEERING PROJECTS | CONTINUED

CMOS Logic-Gate Layout and 8-Bit Adder VLSI Design

Texas State University | VLSI Course Project Fall 2026 | Ongoing

- Designed and simulated transistor-level CMOS logic-gate schematics using LTspice, followed by corresponding IC layouts in Microwind.
- Developing the complete CMOS physical layout of an 8-bit adder in Microwind, integrating individual logic-gate designs into a digital arithmetic circuit.

Simple Processor Design for Integer and Matrix Operations Using HDL

Texas State University | HDL Course Project Fall 2026 | Ongoing

- Developing a simple processor in SystemVerilog, integrating registers, memory, arithmetic logic units (ALUs), and control logic for integer and matrix operations.
- Designing a matrix ALU for 4×4 matrix addition, subtraction, transpose, scalar multiplication, and multiplication using 16-bit elements, alongside integer arithmetic and conditional branching instructions.

Semiconductor Manufacturing and Fabrication Laboratory

Texas State University

- Performed silicon wafer oxidation and oxide-thickness metrology to investigate semiconductor processing and thin-film properties.
- Conducted vacuum-system and PVD-related laboratory investigations, including pump-down and rate-of-rise measurements.
- Practiced photolithography and etching to transfer patterned features onto silicon wafers, developing familiarity with wafer-processing workflows.

Embedded C Programming on Xilinx Zynq Platform

Texas State University

- Developed embedded C applications using memory-mapped I/O to interface with GPIO and seven-segment displays.
- Implemented UART, SPI, and I2C peripheral interfaces, timer-based PWM control, and XADC data acquisition.

TECHNICAL SKILLS

Semiconductor Modeling: Silvaco TCAD (ATLAS), ferroelectric device modeling, I–V and C–V simulation and analysis
VLSI & CMOS Design: Microwind (transistor-level CMOS physical layout), digital logic, Verilog, SystemVerilog, VHDL
Circuit Design & EDA: LTspice; Altium Designer (schematic capture and PCB layout)
Programming & Embedded Systems: C, C++, Python, embedded C, GPIO, UART, SPI, I2C, timers, PWM, XADC
Semiconductor Fabrication: Silicon wafer oxidation, photolithography, etching, PVD fundamentals, oxide-thickness metrology
Laboratory Tools: Oscilloscopes, function generators, digital multimeters, bench power supplies, soldering, hardware prototyping

ENGINEERING & TEACHING EXPERIENCE

Chemistry Laboratory Instructional Assistant Fall 2024 - Spring 2026

Texas State University | Chemistry Department

- Supported undergraduate laboratory instruction, assisting students with experimental procedures, laboratory instrumentation, and safe equipment operation.
- Troubleshot laboratory equipment and provided technical guidance to students during experiments.

ACTIVITIES

- Institute of Electrical and Electronics Engineers (IEEE), Member
- Brain-Engaging Activities
- Exploring New Technologies
- Exploring Nature and Traveling

REFERENCES

Dr. Dinusha Herath Mudiyanselage
Assistant Professor
Ingram School of Engineering
Texas State University
dinusha@txstate.edu

Dr. Mark W. Welker
Associate Professor of Practice
Ingram School of Engineering
Texas State University
mark.welker@txstate.edu

David B. Ownby
Lecturer
Ingram School of Engineering
Texas State University
downby@txstate.edu